

Understanding the factors affecting students' composting attitude and knowledge

Pratheesh Omana Sudhakaran¹, Abu Ahamed Sabbir², Varun Nagaraj³, and Tina Marie Waliczek^{4*}

¹ Department of Agricultural Sciences, Texas State University, 601 University Dr, San Marcos, TX 78666; e-mail: pratheesh81@txstate.edu

² Department of Agricultural Sciences, Texas State University, 601 University Dr. San Marcos, TX 78666; e-mail: pmg92@txstate.edu

³ Round Rock High School, 201 Deep Wood Dr, Round Rock, TX 78681; e-mail: varunnagaraj2009@gmail.com

⁴ Department of Agricultural Sciences, Texas State University, 601 University Dr., San Marcos, TX 78666; e-mail: tc10@txstate.edu

*Correspondence: tc10@txstate.edu ; Tel.: (512) 408- 3635

Abstract

Composting is a critical strategy for reducing food waste and greenhouse gas emissions, yet incorrect sorting of food waste remains a major challenge on college campuses. This study examined how composting knowledge, attitudes, and subjective norms influence students' composting behavior at a large public university. Surveys were administered before and after implementing redesigned composting signage, and 570 valid responses were analyzed using Structural Equation Modeling within the Theory of Planned Behavior framework. Attitudes and subjective norms emerged as the strongest predictors of students' self-reported composting behavior, whereas composting knowledge had no direct effect. The redesigned signage did not significantly alter attitudes, subjective norms, or behaviors, although it was associated with a moderate increase in composting knowledge. The study also found differences in composting knowledge and attitudes by gender and year of study. These findings suggest that efforts to increase campus composting participation should focus on fostering positive attitudes and strong subjective norms rather than relying solely on informational signage.

Keywords: waste management, composting attitudes, composting knowledge, compost, food waste, food recycling

Academic Editor: Firstname Last-name

Received: date

Revised: date

Accepted: date

Published: date

Copyright: © 2026 by the authors. Submitted for possible open access publication under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

1. Introduction

It is estimated that 22 million pounds of food waste is generated on college campuses every year [1,2]. Within these self-contained campus environments, which model other large organizations, there are opportunities to raise awareness of the immediate and long-term effects of food waste disposal in landfills on the environment and influence change [3,4]. Several college campuses and universities across the nation have incorporated interdisciplinary sustainability topics regarding food waste, energy conservation and their environmental impacts into formal or informal curricula [5]. To support these

efforts, some college campuses are integrating composting programs to collect food waste from cafeterias while using the programs to educate students about environmental stewardship [6]. Composting is considered a major food waste reduction method where uneaten food is recycled into a soil resource [7]. For example, Ohio State University is currently home to the largest in-vessel composting system on any college or university campus in the nation collecting food scraps 6 d per week from the Central Food Facility and campus dining halls [8]. The Bobcat Blend composting program at Texas State University diverts tons of food waste annually [3].

The success of a composting program depends on the participants' knowledge and behaviors regarding compostable materials whereby the participants help to reduce the contamination within the flow of feedstocks to the composting process and increase the amount of organic material going toward composting [9]. One of the major factors considered regarding the success of composting programs is the visual prompts and models used to communicate information regarding compostables and composting. Past research found effective signage and placement positively influenced environmental attitudes, knowledge and behavior [10,11]. Sussman et al. [9] suggested that an improved signage would help improve the public perception and behavior about composting.

Educating young adults about environmental issues was shown to have positive effects in research. When people understand the positive implications of their actions by observing results and a tangible product of their labor, future actions are more likely to be positively affected and influence the continuation of positive environmental habits [12,3,13]. Additionally, researchers found that once people started composting, they would continue those behaviors, reducing any further misconceptions about composting [14,15,16].

To design better educational material for enhancing the student's composting behavior, it is critically important to understand the students' attitude and knowledge about composting. Our main goal of this study was to identify the factors that affect the students' composting behavior. To attain this goal, the following objectives were designed.

1. Determine the factors that affect the attitudes of students towards composting. This would help identify who views the process more favorably and is therefore more likely to participate.
2. Determine the knowledge students have about composting. This could provide us with direction to potentially improve composting participation and education, further reducing food waste on college campuses.

2. Results

Summary findings on the SEM analysis for relationships between composting knowledge, attitudes, subjective norms, signage exposure, and composting behavior of students at Texas State University are presented in this section. First, descriptive statistics present an overview of the sample used in the study to lead into the analysis. The results of the SEM are organized by dependent variable: knowledge, attitude, subjective norm, and behavior. Each analysis has provided parameter estimates, standard errors, and significance levels used to measure the direct and indirect relationships stated in the Theory of Planned Behavior model framework. Additional fit indices AIC and BIC have also been provided to check if each analysis has a sufficient degree of model adequacy. As a whole,

these data describe how different psychological and demographic influences can help predict someone's willingness to participate in composting.

2.1. Descriptive Statistics

Table 1. Characteristics of the sample collected from students of Texas State University

during the study of composting attitude and knowledge of a college composting program at Texas State University

Variables	Description	Percentage
Gender	Male	33%
	Female	66%
	Other	1%
Ethnicity	White, not of Hispanic origin	44.40%
	Hispanic	34.10%
	African American	10.10%
	Asian and Pacific Islander	4.20%
	Other	7.20%
Year in School.	Freshman	42%
	Sophomore	21%
	Junior	18%
	Senior	16%
	Graduate Student	3%

After excluding incomplete cases from the original dataset of 603 responses, the structural equation model assessed the relationships among composting knowledge, attitudes, subjective norms, and composting behavior using 570 student responses. Respondents included 306 students who participated in the pre-test survey and 297 who answered the post-test survey. Missing answers were treated as a miscellaneous answer choice "other". More in-depth information about the exact questions asked can be found in Appendix table A.1 in Appendix A.6. Gender identity was disclosed by 570 of those 570 respondents, with 376 (66%) identifying as female, 188 (33%) identifying as male and 6 (1%) identifying as other. Regarding ethnicity, 570 of the respondents self-identified as white (253, 44.4%), 194 (34.1%) as Hispanic, 58 (10.1%) as African American, 41 (7.2%) as other, 24 (4.2%) as Asian or Pacific Islander, and 0 (0%) as American Indian. In reference to their year in school, 570 of the respondents responded, with 239 (42%) of respondents being classified as freshmen, 120 (21%) ranked as sophomores, 103 (18%) juniors participating, 91 (16%) seniors answering, and 17 (3%) ranked as graduate students.

2.2. Knowledge Score Analysis

Table 2. Factors influencing knowledge at university students in Texas State using Structural

Equation Modeling to study the effects of compost attitude, knowledge, subjective norms, and signage on overall composting behavior.

Variables	Estimate	Standard Error	P-Value
Post-Signage	0.250	0.169	0.139
Male	-0.168	0.183	0.358
Native	6.149	1.967	0.002
Asian	0.601	0.447	0.178
Black	1.174	0.289	0.000
Hispanic	1.177	0.188	0.000
Ethnicity - Other	1.044	0.333	0.002
Sophomore	0.111	0.216	0.607
Junior	-0.400	0.239	0.094
Senior	-0.176	0.247	0.475
Graduate Student	0.455	0.644	0.481
Akaike Information Criterion (AIC)			2407.403
Bayesian Information Criterion (BIC)			2463.896
Total Student Responses			570

While analyzing the knowledge scores (table 2), there was not a statistically significant predictor of signage. Native American students may be significant at the 1% level of significance: the knowledge score for Native American students was 6.15 times higher on average than for White students. Black students may be significant at the 1% level of significance; they scored an average of 1.17 times higher than White students. Hispanic students may also be significant at the 1% level of significance at an average of 1.18 times higher than White students. Students who were identified as "Other" may be significant at the 1% level of significance: reporting 1.04 times higher than White students. There was not a significant difference in knowledge for Asian students or males. There were no significant differences in knowledge for school year groups compared to freshmen.

2.3. Attitude Score Analysis

Table 3. Factors influencing attitude at university students in Texas State using Structural

Equation Modeling to study the effects of compost attitude, knowledge, subjective norms, and signage on overall composting behavior.

Variables	Estimate	Standard Error	P-Value
Knowledge Score	-0.010	0.046	0.826
Post Signage	-0.249	0.187	0.182
Male	0.291	0.202	0.150
Native	-1.118	2.187	0.609
Asian	-0.293	0.493	0.552
Black	-0.587	0.324	0.070
Hispanic	-0.672	0.214	0.002
Ethnicity - Other	-0.277	0.370	0.454
Sophomore	0.538	0.238	0.024
Junior	1.275	0.264	0.000
Senior	1.846	0.272	0.000
Graduate Student	0.853	0.711	0.230
Akaike Information Criterion (AIC)			2520.778
Bayesian Information Criterion (BIC)			2581.617
Total Student Responses			570

In the SEM analysis examining predictors of attitudes, neither knowledge nor signage, fixed predictors included in the model, were significant. Hispanic students may significantly affect the attitude score at the 1% level of significance and scored 0.67 times lower than White students. Black students may also affect their attitude score at the 10% level of significance and scored 0.59 times lower than White students. Students who are older than the baseline sample may have reported more favorable attitudes towards composting. For example, sophomores may affect the attitude score at the 5% level of significance and score 0.54 times higher than freshman students. For juniors and seniors, they may affect attitudes at the 1% level of significance and scored 1.28 and 1.85 times higher

on average, respectively, compared to freshman. Graduate students did not indicate any significant difference in score when examining attitudes.

Additionally, upper-level students (sophomores to seniors) had more positive attitudes toward composting relative to freshmen, which suggests that students' experiences on campus and exposure to campus sustainability programs may be related to improvements in perception of composting. Gender differences presented as small, with women having slightly higher knowledge, and no statistically significant differences regarding attitude.

2.4. Subjective Norms Score Analysis

Table 4. Factors influencing subjective norms at university students in Texas State using Structural Equation Modeling to study the effects of compost attitude, knowledge, subjective norms, and signage on overall composting behavior.

Variables	Estimate	Standard Error	P-Value
Knowledge Score	0.043	0.030	0.152
Attitude Score	0.256	0.027	0.000
Post Signage	-0.185	0.122	0.131
Male	-0.031	0.132	0.812
Native	0.803	1.430	0.574
Asian	0.475	0.323	0.141
Black	0.065	0.212	0.760
Hispanic	-0.035	0.141	0.802
Ethnicity - Other	-0.094	0.242	0.697
Sophomore	0.146	0.157	0.350
Junior	0.126	0.176	0.475
Senior	0.155	0.185	0.401
Graduate Student	-0.044	0.465	-0.004

Akaike Information Criterion (AIC)	2038.243
Bayesian Information Criterion (BIC)	2103.428
Total Student Responses	570

In the SEM analysis of subjective norms, attitudes were a strong and significant predictor of subjective norms, meaning that as students had a more positive attitude toward composting, they were more likely to see themselves as living in a climate where their peers supported composting, as shown by the strength of the relationship that was statistically significant at the 1% level. The result showed that knowledge and signage exposure, as well as gender, ethnicity and school year of the subjects studied, were not statistically significant predictors of subjective norms, and thus, students from all demographic groups perceived almost the same level of social support and encouragement to compost because of the institutional climate surrounding composting. The result is suggesting that the best way to strengthen students' perceptions of social support for composting was to strengthen students' attitude toward composting which could help build actual perceptions of social support for composting.

2.5. Behavior Score Analysis

Table 5. Factors influencing composting behavior at university students in Texas State using Structural Equation Modeling to study the effects of compost attitude, knowledge, subjective norms, and signage on overall composting behavior.

Variables	Estimate	Standard Error	P-Value
Attitude Score	0.094	0.012	0.000
Knowledge Score	0.017	0.013	0.191
Subjective Norm Score	0.121	0.018	0.000
Post Signage	0.032	0.051	0.534
Male	-0.083	0.056	0.133
Native	0.440	0.600	0.463
Asian	-0.068	0.136	0.615
Black	0.016	0.089	0.855
Hispanic	0.052	0.059	0.036

Ethnicity - Other	0.013	0.102	0.005
Sophomore	0.139	0.066	0.083
Junior	0.028	0.074	0.016
Senior	0.093	0.078	0.050
Graduate Student	0.096	0.195	0.018
Akaike Information Criterion(AIC)			1049.820
Bayesian Information Criterion(BIC)			1119.350
Total Student Responses			570

When predicting composting behavior, both attitudes and subjective norms may affect the outcome at the 1% level of significance. Students with more positive attitudes or stronger social norms were more likely to engage in composting behaviors. Knowledge and signage were not significant predictors of behavior. Most demographic variables also did not show a significant influence, although sophomores may affect behavior at the 5% significance level and scored 0.14 times higher than freshmen.

3. Discussion

Attitudes and perceived social norms were the main predictors of composting behavior. Knowledge did not, by itself, drive behavior. This aligns with the Theory of Planned Behavior [17], which states that attitudes that originated internally, and social norms that appeared externally, combine to form behavioral intention. Those students who perceived composting behavior as socially expected and of value personally, were more likely to engage in the behavior, regardless of signage exposure or any objective understanding of the factual basis for composting.

These findings present an important implication for campus sustainability initiatives — in some cases, educational and social interventions may be better than a static signage production. While clear and available signage is still very valuable to operational clarity, it appears that its potential influence is minimal until some other intervention(s) takes place, such as peer-led education, incorporation into curriculum/classroom education, and public appeals to increase salience of pro-environmental social norms. In short, reinforcing a culture of composting on campus will take more than visual reminders; it will take social modeling, social feedback and repeated opportunities for staff and students to engage, to enhance the strength of attitudes and subjective norms over time.

Nonetheless, there are some limitations that bear consideration when interpreting the study findings. The post-signage measure period was relatively short (two weeks) and possibly not sufficient time for the new signage to adequately influence behavior significantly. A longer duration could have easily been needed to detect any delayed effect. Furthermore, the observational design of the study did not allow us to track individual students across time, leaving uncertainty regarding whether the same students were engaged in both pre- and post-testing. This diminishes our ability to account for individual-level

variation. Additionally, we had a limited ability to ensure the student population was entirely comparable pre- and post-signage implementation. We also relied on self-reported behavior rather than direct observation; although self-reported data are more efficient to collect, they could be biased. Furthermore, we did not account for possible non-linear relationships or interaction effects that may become evident in a larger or more diverse dataset.

Future studies ought to examine the significance of peer networks, social identity, and feedback loops in influencing composting involvement in varied campus settings. It would also be beneficial to extend this work beyond the university context — to workplaces, international airports or other public spaces — to understand how social context moderates the transferability of the design of information. Ultimately, while signage may raise awareness, sustained participation in composting is about developing collective norms and long-term educational engagement that are consistent with students' values and lived experiences.

4. Materials and Methods

4.1. Composting Program

At Texas State University, the Bobcat Blend Composting project began in 2008. When the project was initiated, cafeteria waste from two dining halls and invasive plant species from the San Marcos River, which runs through campus, were the original feedstocks used in compost piles. However, the project has expanded to include collections from five of six cafeterias. Additionally, the project collects leaves and tree trimmings from the Grounds and the Department of Agricultural Science, coffee grounds from the Honors Department, food waste from Nutrition Labs and some special events.

Since the program collects pre- and post-consumer organic waste from cafeterias, students are actively involved in sorting the post-consumer food waste into the proper bins. Signage directs students and helps them make decisions. The compostable materials are taken to the compost site, 10 miles from campus, and processed in traditional windrow piles. Compost is used back on campus in the greenhouses and gardens and sold to the community.

4.2. Survey questions, instrument and scoring

The survey was modeled using instruments which are known to be both reliable and valid [11,2]. The survey questions contained five sections (compost knowledge, compost attitudes, subjective norms, behavior and demographics) and are outlined in detail with scoring and coding information below.

4.2.1. Compost knowledge instrument and scoring.

Composting knowledge was assessed through 20 true-false questions which assessed knowledge of basic composting concepts. Some sample items for true-false statements were "The nutrients in food can be recycled," "Paper is biodegradable" and "The product of a compost pile is similar to topsoil." Each correct answer was awarded 2 points, whereas incorrect/incomplete answers were awarded 1 point, meaning the total possible score was between 20-40 points and higher scores equated to greater knowledge about composting, exact questions can be found in Appendix A.2. The goal of this scoring system was to lessen the negative impact of an incorrect answer while avoiding 0 values, which could introduce issues while modeling. At the same time having different total scores represents a more or less accurate understanding of composting. Prior research reported an acceptable internal reliability with this scale, with a Cronbach's alpha of 0.750 [3], which is acceptable reliability for studies rooted in educational and behavioral research [17].

4.2.2. Composting attitudes, instruments and scoring.

Students reporting an awareness of the composting program at the university were asked six questions regarding their own attitudes towards composting. Examples of survey questions included, "when I sort my food waste in the cafeteria/dining hall, I understand what can be composted," "I know what happens to the food sorted into campus dining hall/cafeteria compost bins," and "after the meal if I have leftover food, I save it to eat later," Likert-type scale responses and scores ranged from "never" (scoring 1 point) "sometimes/usually" (scoring 2 points) and "always" (scoring 3 points). A maximum score of 18 points was possible for this section of the survey with higher scores indicating more positive composting attitudes. The questions asked to determine composting attitude can be found in Appendix A.3. This section of the survey had a Cronbach's reliability coefficient of 0.81 indicating a suitable level of reliability for this type of research [18].

4.2.3. Composting subjective norms, instruments and scoring.

Perceived social expectations and peer behaviors related to composting were assessed through four survey items asking students to rate their agreement with statements about how their university expects them to separate compost from non-compost and how their friends throw the compost materials correctly into the compost bin. Responses were scored on a 3-point Likert-type scale (strongly disagree [1], strongly agree [3]) and summed to create a total score (4-12) where a higher total score indicates a more positive attitude toward composting. The exact questions used to determine subjective norms can be found in Appendix A.4. The perceived social norms construct identified in this study supports previous researchers' findings that perceived social norms and peer behaviors affect pro-environmental behavior [17,26]; therefore, the findings are relevant to other pro-environmental programs, especially on college campuses.

4.2.4. Composting behavior, instruments and scoring.

To assess people's composting habits and their recycling knowledge, five questions regarding the use of compost bins at home and on campus were included. Participants were asked about how frequently they composted on campus and at home, how accurately food waste was sorted into compost, recycle or trash bins, and how people corrected their mistakes of placing plastic into the compost bin. Responses were scored on a numerical scale from 1 (doesn't participate in activity) to 3 (always participates in activity). The exact questions used to find composting behavior can be found in Appendix A.5. Scores between 5 and 15 indicated the frequency with which an individual participated in composting, the higher the score, the more frequently that individual composted. The total score gives an individual a general view of their composting habits, as well as how effectively they are correcting mistakes made when disposing of items in compost. Researchers studying environmental behavior utilize this type of questioning to differentiate an individual's composting practice from their knowledge about recycling or composting and their feelings toward both practices.

4.2.5. Demographic section.

Demographic data was collected including gender, ethnicity and year in school to make comparisons amongst different participating groups. Gender was separated into three categories: male, female, and other. Ethnicity was divided into six sections: American Indian or Alaskan Native; Asian or Pacific Islander; African American; Hispanic; White, not of Hispanic origin; and other. Years of classification in school included the categories of freshman, sophomore, junior, senior, senior with more than 4 years in school, and graduate student. The exact questions used to determine the demographic values can

be found in Appendix Table A.1 in Appendix A.6 focused on discussing the exact demographic questions asked.

4.3. Study sites and survey administration.

All dining halls incorporating Bobcat Blend, the campus composting program, were included in the study. Considering the findings from Sussman et al. [9], this survey considered the quality of signage as one of the factors affecting composting attitude and knowledge. Thus, data was collected before (hereafter referred to as pre-test) and after (referred to as post-test) new signage was implemented. Three campus buildings where post-consumer waste was collected by Bobcat Blend were also included in the study. Both pre-test and post-test surveys were administered in dining halls, classrooms, and public areas within the study sites. Students were approached and asked to participate in this study and incentives were given after the completion of each survey. Incentives included Bobcat Horticulture pens, flashlights, and stickers, along with a chance of entering a drawing for a \$25 gift card. Pre-test and post-test surveys were administered at study sites at various times of the day to ensure an overall representative sample of the campus population at the time. All surveys were administered by one researcher for consistency of administration. Pre-test surveys were administered with older educational signage in place. Post-test surveys were administered after the newly designed signage had been in place for two weeks.

4.4. Pre-test signage.

The original signage for the composting program utilized actual photographs of products and packaging to help individuals understand items that could be composted, items that could be recycled, and the items in neither category that would be landfilled. This existing signage was based on research and examples from successful programs [19]. A similar, successful program with an emphasis on signage was implemented from 2014 to 2016 by the University of Michigan alongside efforts to promote composting knowledge [20]. Another study emphasized the influence of signage and other posted notifications on the environmental actions of students [21].

4.5. Post-test signage and treatment.

The newly designed signage and educational outreach included generic, very simple and easy-to-read line drawings with labels of compostable and recyclable items and those that should go to the landfill. New signage was created by a graphic designer and based on professional experience, recommendations, and the latest information regarding education in the field at the time of the study. Obtrusive or prominently displayed signage has been shown to increase compliance [22]. Therefore, signage was posted in highly visible locations. Informational posters and table tents advertising the Bobcat Blend program and site were also posted at the dining halls and locations where compostables are collected to help educate individuals on the Bobcat Blend program and composting on campus.

New signage was in place for two weeks. This length of time was chosen because it was the maximum amount of time the dining halls would allow for table tents to be posted. However, other educational signage remained in place in approved locations such as bulletin boards after post-test surveys were administered.

4.6. Data analysis.

All the surveys were collected and entered an Excel (Microsoft, Redmond, WA) spreadsheet, which was converted into a file in SPSS™ (version 20.0; IBM Corp., Armonk,

NY). Descriptive and frequency tests were used to gain an overview of the data. Structural equation modeling was used to examine the relationships among multiple variables simultaneously. This study aims to determine the factors affecting composting behavior and therefore a Structural Equation Model (SEM) was created. The dependent variables considered for the model was behavior. One of the main factors considered for the model was the signage. The signage variable is represented as a dummy variable with 1 being new signage and 0 being the old signage. The other independent factors considered were gender, year in school and ethnicity which were included as categorical variables as well as the subjective norms and a student's knowledge. Since attitude could have an influence on knowledge, attitude scores were also included in the model.

4.7. Theoretical/Conceptual Framework.

In this research study, we employed the Theory of Planned Behavior (TPB) to consider student behavior regarding composting. In the Theory of Planned Behavior, behavior is determined by attitude, subjective norms, and perceived behavioral control [23]. Although composting knowledge is not a formal construct within the TPB, this study operationalized composting knowledge as a functional substitute for perceived behavioral control, based on the premise that greater knowledge of composting practices reflects a higher perceived ability to perform the behavior [24]. Previous research has identified this connection between environmental knowledge and attitudes - specifically environmental knowledge being related to more favorable attitudes and greater behavioral efficacy [24]. This study addresses two research objectives:

1. To examine important factors influencing student attitudes towards composting.
2. To measure levels of composting knowledge to inform future educational outreach opportunities on college campuses.

The Theory of Planned Behavior (TPB) provides a robust foundation for modeling composting behavior as a function of attitudes, knowledge, and subjective norms. A recursive path model, in which knowledge influences attitudes and subjective norms act as antecedents of behavior, is consistent with prior environmental behavior research. Taylor applied a TPB framework in the context of household waste reduction and found that attitudes, norms, and control beliefs were critical determinants of behavioral outcomes [25].

More recently, Oehman et al. modeled household food-waste separation in New York State and demonstrated that attitudes, norms, and knowledge strongly predicted behavioral intent while also highlighting contextual barriers that moderated these predictors [26]. In a broader sustainability context, García-Salirrosas et al. found that environmental awareness significantly shaped both attitudes and knowledge which together predicted willingness to pay for green products in Peru [27]. Collectively, this literature provides the empirical foundation for specifying direct and indirect pathways in which composting knowledge and attitudes influence subjective norms, and in which attitudes, knowledge, and subjective norms jointly predict composting behavior.

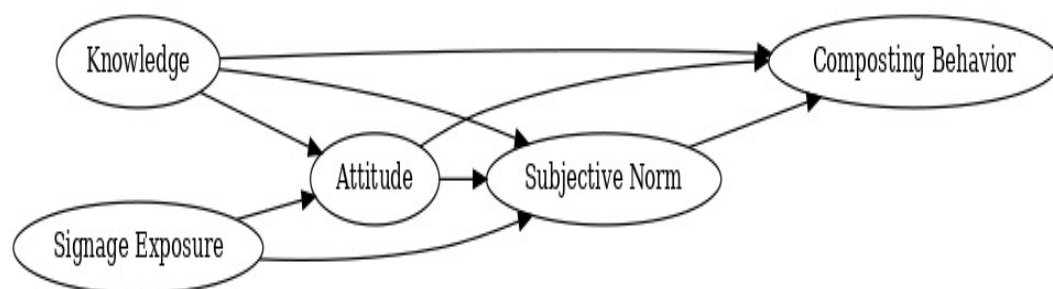


Figure 1. Conceptual model illustrating the hypothesized relationships among signage exposure, composting knowledge, attitudes, subjective norms, and composting behavior based on the Theory of Planned Behavior framework.

The diagram provided illustrates how signage exposure and knowledge lead to attitude and subjective norm, which ultimately influences composting behavior. Our analysis will determine if the relationships between signage exposure and knowledge, attitude and subjective norm, and composting behavior hold true to the students' data collected.

Guided by this precedent, we constructed a TPB-aligned modeling framework using observed variables available in our student survey dataset. Composting behavior (B), the dependent variable, was measured with questions such as: "In your usual day, do you compost?", along with other questions meant to assess a person's composting activities. Attitudes (A) were operationalized through a composite score of six Likert-scale items capturing students' confidence, understanding, and engagement with composting practices. Knowledge Score (K) was estimated using a knowledge score based on twenty true/false items assessing familiarity with composting. Subjective norms (N) were measured as the average of four Likert-scale items reflecting awareness of a campus composting program and perceived social expectations related to composting. Demographic variables (gender, year in school, and ethnicity) and signage treatment (pre-test vs. post-test) were included as controls. This design directly informed the specification of our structural equation model (SEM), in which composting behavior was modeled as a function of attitudes, subjective norms, and knowledge, alongside signage and demographic controls. In doing so, our methodology reflects both the theoretical foundation of the TPB and the practical constraints of SEM estimation, resulting in a model that is grounded in established research while tailored to our dataset.

4.8. Statistical Analysis.

This research employed a structural equation model to investigate the influence of various psychological and demographic variables on students' composting behavior. The model was guided by theories associated with the Theory of Planned Behavior. Typically, individual actions will be influenced by some internal attitude about engaging in the behavior, some social expectation from others (or subjective norm), and some knowledge about engaging in the behavior. For the model, attitude was defined as a student's attitude about composting. For example, do they think it is good, important, or beneficial? The subjective norm refers to the social pressure the student perceives to perform an action, that is, if friends, peers, and the campus community expect them to perform said action. The behavior is their composting behavior, for example, do they separate food waste into compost bins? Knowledge was also included in the model (or whether students knew what composting was and how to do it correctly.) The model also tested signage exposure, having students see new graphic composting signs. Collectively, the model aimed to assess whether knowledge and/or exposure to signage directly affects composting behavior, or whether there is a mediating effect of knowledge or signage to influence changes in attitudes and social norms.

4.9. Descriptive Analysis.

Through the use of descriptive statistics, an overview of various demographic characteristics of the study sample such as composting knowledge, attitude, subjective norms, and composting behavior was created. Frequency and percentage distributions for several categorical variables (gender, race, year of university, and signage treatment) was also derived from a student's responses on surveys (the study conducted surveys at pre (before) and post (after) the treatment). In addition, measures of central tendency (mean) and

dispersion (standard deviation) were calculated for continuous composite variables including knowledge score, the attitudes score, social norms score, and from the behaviors score. Using such descriptive analysis, comparisons of patterns across similar demographic groups and survey conditions were modeled and interpreted.

4.10. Structural Equation Modeling.

The structural model was delineated as a system of linear equations. Composting knowledge was, in the first instance, modeled as a function of demographic variables and exposure to signage, as this equation:

$$K_i = \delta_0 + \delta_1 \text{Signage}_i + \delta_2 \text{Gender}_i + \delta_3 \text{Year}_i + \delta_4 \text{Ethnicity}_i + \epsilon_{K_i}.$$

Next, composting attitude was modeled as a function of knowledge and demographic factors:

$$A_i = \beta_0 + \beta_1 K_i + \beta_2 \text{Signage}_i + \beta_3 \text{Gender}_i + \beta_4 \text{Year}_i + \beta_5 \text{Ethnicity}_i + \epsilon_{A_i}.$$

Subjective norms were modeled as a function of demographic variables and signage treatment:

$$N_i = \gamma_0 + \gamma_1 K_i + \gamma_2 A_i + \gamma_3 \text{Signage}_i + \gamma_4 \text{Gender}_i + \gamma_5 \text{Year}_i + \gamma_6 \text{Ethnicity}_i + \epsilon_{N_i}.$$

Finally, composting behavior was modeled as a function of all three TPB constructs—attitude, knowledge, and subjective norms—along with relevant demographic and treatment controls:

$$B_i = \alpha_0 + \alpha_1 A_i + \alpha_2 K_i + \alpha_3 N_i + \alpha_4 \text{Signage}_i + \alpha_5 \text{Gender}_i + \alpha_6 \text{Year}_i + \alpha_7 \text{Ethnicity}_i + \epsilon_{B_i}$$

where B_i is the self-reported composting behavior for student i , A_i is the attitude score, K_i is the composting knowledge score, and N_i is the composite subjective norm score. The variable *Signage* indicates whether a student was in the pre-test or post-test condition (0 = old signage, 1 = new signage) and demographic controls are gender, year in school, and ethnicity. This modeling framework allows us to test the primary hypothesis of TPB—that internal beliefs (attitude and perceived control, operationalized as knowledge) and external social expectations (norms) predict behavior—while at the same time controlling for demographic influences and environmental intervention (signage). By basing the analysis on theory and associating each construct with survey items that were widely available, the model provides a strong theory-based framework for understanding student composting participation on campus.

5. Conclusions

The aim of the current study was to investigate the motivation of college students to engage in composting using the Theory of Planned Behavior. It was found that students were more likely to compost if they valued composting personally and believed that those around them would expect, as well as encourage, them to compost. On the other hand, while knowledge of composting and exposure to new signs to promote composting are both important, these factors alone are not sufficient to change students' composting behaviors.

Although the level of knowledge regarding composting did differ among students and some student attitudes were related to their level of composting knowledge,

knowledge of composting does not necessarily lead to an increase in actual composting behaviors. Students that had a more positive attitude regarding composting also tended to perceive greater social support than those that held a less positive attitude. The social norms associated with composting were the greatest predictors of whether students would engage in composting behaviors. The results of this study illustrate the importance of the social environment on a college campus and show that a student's behaviors regarding sustainability are strongly influenced by the norms and expectations of their peers.

The findings presented in this article support a conclusion that improved signage alone does not produce a large effect and indicates that visual prompts may not sufficiently influence student behavior for an extended period. While improved signage is helpful for educating students on how to sort their waste correctly to minimize contamination, all forms of signage are most effective when used in conjunction with more comprehensive strategies that enhance students' perception of composting as a common and valued behavior on college campuses. Peer-to-peer engagement, classroom integration, and visible institutional support are examples of potential strategies that may increase the likelihood of long-term participation in composting activities.

Based on the above findings, creating an environment that encourages composting in colleges requires an approach that goes beyond increasing awareness and providing instructions. For sustainable behavioral change to be successfully implemented, students need to feel personally connected to composting, see their behaviors being endorsed by their peers and communities, and perceive that participating in composting activities is an important component of being environmentally responsible. Future research should investigate the use of longer-term engagement strategies in conjunction with more interactive methods of teaching sustainability to gain a better understanding of how attitudes about composting develop and how students' perceptions of social norms change over time. Ultimately, the most effective method of reducing the amount of food waste generated on college campuses and increasing the development of environmentally responsible behavior among students may be to foster a culture that supports composting.

Author Contributions: Dr. Waliczek: Survey design, data collection; Dr. Omana Sudhakaran: Manuscript preparation, data analysis; Mr. Sabbir: Data Analysis, manuscript preparation; Mr. Nagaraj (High School student): was an intern during last summer and did most of the data analysis and initial draft of the manuscript. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding authors.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

TPB	Theory of Planned Behavior
SEM	Structural Equation Modeling
AIC	Akaike Information Criterion
BIC	Bayesian Information Criterion

Appendix A: Survey Instrument and Coding

Appendix A.1

The questions asked to gauge compost knowledge, attitude, subjective norms, composting behavior, and demographic variables have been given below for any survey reproduction. Null/Blank responses were given the average of all the other responses in the row.

Appendix A.2. Compost Knowledge Questions

There were 20 (True/False) questions asked to estimate participant's compost knowledge. Correct answers were given 2 point whereas incorrect answers were given 1 points. Total points were added up to get the knowledge score. The questions asked are below:

1. Paper is biodegradable
2. The nutrients in food can be recycled
3. Plastic wrap should go into the compost
4. Changing food into soil takes several weeks or months
5. Soil is made out of organic material and sand
6. Worms are decomposers
7. Decomposers in a compost pile need oxygen
8. It doesn't matter if the compost pile is turned as long as it is wet
9. The product of a compost pile is similar to topsoil
10. The best way to get rid of leaves is to put them in a plastic bag for trash pick-up
11. When compost gets compacted, it can cause unpleasant odors
12. The center of a working compost pile is very cool
13. Compost piles are always very smelly
14. Nature recycles
15. People who live in apartments can't compost
16. Composting is one way to recycle
17. Decomposers need only food and air to do their work in a compost pile
18. Yard scraps such as grass clippings and leaves can be composted
19. Composting works only if it is done in a specially designed compost bin
20. Usually, animals are not attracted to a compost bin if meat and bones are not put into it

Appendix A.3. Compost Attitude Questions

There were 6 Likert scale questions asked to estimate participant's attitude towards composting. There were 3 possible values: 1 for "never," 2 for "sometimes/usually," and 3 for "always". Total points were added up to get the attitude score. The questions asked are below:

1. I understand what composting is
2. When I sort my food waste in the cafeteria/dining hall, I understand what can be composted
3. I know what happens to the food sorted into campus dining hall/cafeteria compost bins
4. After the meal if I have leftover food, I save it to eat later.
5. The product from composting food waste, compost, can be useful.
6. How do you feel about having a campus composting program

Appendix A.4. Subjective Norms Questions

There were 4 Likert scale questions asked to estimate participant's surrounding atmosphere towards composting. There were 3 possible values: 1 for "never," 2 for "sometimes/usually," and 3 for "always". Total points were added up to get the subjective norms score. The questions asked are below:

1. My school requests that food and other compostable wastes be separated from non-compostable wastes.
2. My friends are careful not to throw plastic utensils into the compost food bin in the campus dining hall/cafeteria
3. My friends sort their meal waste correctly
4. I notice when other students sort their meal waste correctly

Appendix A.5. Behavior Questions

There were 5 Likert scale questions asked to estimate participant's behavior towards composting. There were 3 possible values: 1 for "never," 2 for "sometimes/usually," and 3 for "always". Total points were added up to get the overall behavior score. The questions asked are below:

1. I compost on campus
2. I compost at home/in my residence hall
3. I sort my meal waste carefully into the campus dining hall/cafeteria compost, recycling, and trash bins
4. If plastic accidentally lands in the food waste bin I remove it
5. In your usual day, do you compost

Appendix A.6. Demographic Questions

There were 6 demographics questions asked to gain insight about various groups being studied. If Null or incomplete responses were found we replaced it with an "other" or "neutral" option. The scoring and questions are below.

Table A1. Demographic questions and scoring.

Demographic Trait	Unique Values Count	Possible Values
Gender Identification	Female; Male; Other	3
What is your ethnicity	American Indian or Alaskan Native; Asian or Pacific Islander; Black, not of Hispanic origin; Hispanic; White, not of Hispanic origin; Other	6
What is your year in school	Freshman; Sophomore; Junior; Senior; Graduate; Non-response	6
Rate your overall general knowledge of compost	Poor; Fair; Good; Very Good; Excellent	5
If you had a composting program on campus, do you think you would be aware of it	No; Yes	2

How do you feel about having a campus composting program	Strongly negative; Negative; Neutral; Positive; Strongly positive	5
--	---	---

References

- National Public Radio. When food is too good to waste, college kids pick up the scraps. Available online: <https://www.npr.org/sections/thesalt/2015/02/27/389284061/when-food-is-too-good-to-waste-college-kids-pick-up-the-scraps> (accessed on 27 February 2015).
- Waliczek, T.C.; McFarland, A.; Holmes, M. The relationship between a campus composting program and environmental attitudes, environmental locus of control, compost knowledge, and compost attitudes of college students. *HortTechnology* 2016, 26, 592–598. <https://doi.org/10.21273/HORTTECH03320-16>
- Sanders, J.; Waliczek, T.M.; Gandonou, J.M. An economic analysis of a university educational cafeteria composting program—Bobcat Blend. *HortTechnology* 2011, 21, 639–646. <https://doi.org/10.21273/HORTTECH.21.5.639>
- Lilly, A. Colleges reducing food waste and greening the earth – How institutions are finding resourceful, sustainable solutions. Available online: <http://www.universitybusiness.com/article/colleges-reducing-food-waste-and-greening-earth> (accessed on 12 January 2016).
- University of Texas at Arlington. UTA launches new institute for sustainability and global impact. Available online: <https://www.uta.edu/enews/mavwire/2015/02/05.php> (accessed on 5 February 2015).
- University of Georgia Extension. Food waste composting: Institutional and industrial application. Available online: <https://extension.uga.edu/publications/detail.html?number=B1189&title=Food%20Waste%20Composting:%20Institutional%20and%20Industrial%20Application> (accessed on 1 January 2020).
- Ohio State University. Compost. Available online: <https://www.ohio.edu/facilities/grounds/compost> (accessed on 1 January 2021).
- Sussman, R.; Greeno, M.; Gifford, R.; Scannell, L. The effectiveness of models and prompts on waste diversion: A field experiment on composting by cafeteria patrons. *J. Appl. Soc. Psychol.* 2013, 43, 24–34. <https://doi.org/10.1111/j.1559-1816.2012.00978.x>
- Arcury, T.A. Environmental attitude and environmental knowledge. *Hum. Organ.* 1990, 49, 300–304.
- Mosler, H.-J.; Tamas, A.; Tobias, R.; Rodríguez, T.C.; Miranda, O.G. Deriving interventions on the basis of factors influencing behavioral intentions for waste recycling, composting, and reuse in Cuba. *Environ. Behav.* 2008, 40, 522–544. <https://doi.org/10.1177/0013916507300114>
- Aguilar, O.M.; Waliczek, T.M.; Zajicek, Z.M. Growing environmental stewards: The overall effect of a school gardening program on environmental attitudes and environmental locus of control of different demographic groups of elementary school children. *HortTechnology* 2008, 18, 243–249. <https://doi.org/10.21273/HORTTECH.18.2.243>
- Ramsey, C.E.; Rickson, R.E. Environmental knowledge and attitudes. *J. Environ. Educ.* 1976, 8, 10–18. <https://doi.org/10.1080/00958964.1976.9941552>
- Bagozzi, R.P.; Baumgartner, H.; Yi, Y. State versus action orientation and the theory of reasoned action: An application to coupon usage. *J. Consum. Res.* 1992, 18, 505–518. <https://doi.org/10.1086/209277>
- Boldero, J. The prediction of household recycling of newspapers: The role of attitudes, intentions and situational factors. *J. Appl. Soc. Psychol.* 1995, 25, 440–462. <https://doi.org/10.1111/j.1559-1816.1995.tb01598.x>
- Dahab, D.J.; Gentry, J.W.; Su, W. New ways to reach non-recyclers: An extension of the model of reasoned action to recycling behaviors. *Adv. Consum. Res.* 1995, 22, 251–256.
- Ajzen, I. The theory of planned behavior. *Organ. Behav. Hum. Decis. Process.* 1991, 50, 179–211. [https://doi.org/10.1016/0749-5978\(91\)90020-T](https://doi.org/10.1016/0749-5978(91)90020-T)
- Gall, M.D.; Borg, W.R.; Gall, J.P. *Educational Research: An Introduction*, 8th ed.; Allyn and Bacon: White Plains, NY, USA, 2006.
- Sussman, R.; Gifford, R. Be the change you want to see: Modeling food composting in public places. *Environ. Behav.* 2011, 45, 323–343. <https://doi.org/10.1177/0013916511431274>
- School for Environment and Sustainability. Composting. Available online: <https://seas.umich.edu/about/composting> (accessed on 1 January 2016).
- Dixon, R.S.; Knott, T.; Rowsell, H.; Sheldon, L.; Moore, D.W. Prompts and posted feedback: In search of an effective method of litter control. *Behav. Change* 1992, 9, 2–7. <https://doi.org/10.1017/S0813483900006495>
- Aronson, E.; O’Leary, M. The relative effectiveness of models and prompts on energy conservation: A field experiment in a shower room. *J. Environ. Syst.* 1982, 12, 219–224. <https://doi.org/10.2190/UBD5-4Y9B-61EF-WUM6>

22. Sansom, R. Theory of Planned Behavior. In Change Theories Collection; ASCN, 2021. Available online: https://ascnhighered.org/ASCN/change_theories/collection/planned_behavior.html (accessed on 1 January 2021). 626
23. Bradley, J.; Waliczek, T.M.; Zajicek, J.M. Relationship between environmental knowledge and attitudes of high school students. 627
J. Environ. Educ. 1999, 30, 17–21. <https://doi.org/10.1080/00958969909601873> 628
24. Taylor, S.; Todd, P. Understanding household garbage reduction behavior: A test of an integrated model. *J. Public Policy Mark.* 630
1995, 14, 192–204. <https://doi.org/10.2307/30000128> 631
25. Oehman, J.M.; Luo, H.; Harman, B.; Toland, M. What predicts and prevents source separation of household food waste? An 632
application of the Theory of Planned Behavior. *Resour. Conserv. Recycl.* 2022, 186, 106492. <https://doi.org/10.1016/j.resconrec.2022.106492> 633
26. García-Salirrosas, E.E.; Millones-Liza, D.Y.; Álvarez-Risco, A.; Yáñez, J.A.; Del-Aguila-Arcatales, S. Influence of environmental 635
awareness on the willingness to pay for green products: An analysis under the application of the Theory of Planned Behavior 636
in the Peruvian market. *Front. Psychol.* 2024, 14, 1282383. <https://doi.org/10.3389/fpsyg.2023.1282383> 637

638

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual 639
author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any 640
injury to people or property resulting from any ideas, methods, instructions or products referred to in the content. 641

642